

Day 5 — 3-Digit $\times$ 3-Digit Near 1000

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 997×996 mentally.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Extension to base 1000.** Same rule, base now 1000.
 - $997 \rightarrow$ deficit 3.
 - $996 \rightarrow$ deficit 4.
 - Left: $997 - 4 = 993$.
 - Right: $3 \times 4 = 12$. Must be 3 digits (because base has 3 zeros). Pad: **012**.
 - Combine: **993012**.
2. **Above 1000.** $1003 \times 1004 = 1007 \mid 012 = \mathbf{1007012}$.
3. **Mixed case.** $1003 \times 997 = (1000+3)(1000-3) = 10^6 - 9 = 999991$. (Difference of squares.)
4. **Pattern recognition.** The base extends naturally: 10, 100, 1000, 10000... Each next-decade-base just shifts everything. The CORE rule (cross-subtract + product-of-deficits) is identical.
5. **Independent practice.** 5 near-1000 problems. The challenge: keeping the right-part digit count correct.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces 3-Digit $\times$ 3-Digit Near 1000.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).